



Mechanisms for electrochemical performance enhancement by the salt-type electrolyte additive, lithium difluoro(oxalato)borate, in high-voltage lithium-ion batteries

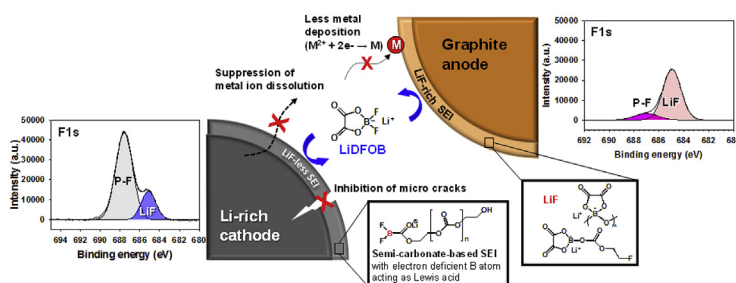
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HIGHLIGHTS

- Lithium difluoro(oxalato)borate (LiDFOB) is used as an electrolyte additive.
- LiDFOB affects both cathode and anode solid electrolyte interphase (SEI) formation.
- LiDFOB produces a semi-carbonate-based SEI on Li-rich cathodes.
- The LiDFOB-derived SEI on graphite anodes consists mostly of LiF.
- Excellent electrode performance is achieved with LiDFOB.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 3 March 2017
Received in revised form
24 April 2017
Accepted 26 April 2017

Keywords:

Lithium-ion battery
LiPF₆-based electrolyte
Graphite anodes
Li-rich cathodes
Lithium difluoro(oxalato)borate
Solid electrolyte interphase

ABSTRACT

Lithium difluoro(oxalato)borate (LiDFOB) with one oxalate moiety bonded to a central boron core was employed as a salt-type additive to enhance the interfacial stabilities of high-voltage Li-rich cathodes and graphite anodes. Our investigation revealed that the LiDFOB additive modified the surface film on the electrodes and effectively restrained degradation of the cycling performance of the electrodes. Investigation of the surface chemistries of the electrodes confirmed that LiDFOB produces a LiF-less surface film on the Li-rich cathode and a LiF-rich surface film on the graphite anode. Moreover, the use of 1% LiDFOB drastically improved the rate capabilities of Li-rich cathodes and graphite anodes. Within 100 cycles at a rate of C/2 at 25 °C, only 45.8% of the initial discharge capacity of a high-voltage Li-rich/graphite full cell was delivered in the baseline electrolyte, while the LiDFOB-containing electrolyte retained 82.7%.

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1. Introduction

Lithium-ion battery (LIB) technology has grown rapidly with advancements in the development of portable electronic devices. However, improvement of the energy density in LIBs is critical

toward their successful application in electric vehicles (EVs) [1–4]. One of the most feasible ways to increase the energy densities of these batteries is to develop electrode materials with high reversible capacities. The cathode material used in commercialized LIBs, LiCoO₂, is not suitable for EVs and portable electronic devices requiring high energy density because of its unsatisfactory capacity (~160 mAh g⁻¹) [1,4,5]. In recent years, Li-rich cathodes (xLi₂MO₃-(1-x)LiMO₂ (M = Co, Mn, Ni)) have attracted attention because they can deliver high capacities that exceed 200 mAh g⁻¹ and enable

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high operating voltages that increase the energy densities of the batteries [6,7]. Despite the appreciable advantages of Li-rich cathodes, their application in LIBs is limited due to several major challenges for their implementation. O_2 gas, evolved from the activation of the Li_2MnO_3 phase during the initial charge process, may be converted to highly reactive oxygen radicals and catalytically accelerate unwanted electrolyte decomposition at the cathode [8,9]. Accordingly, the anodic stability of the electrolyte at high voltages may be reduced further [10]. In addition, Li-rich cathodes do not retain their high capacities on prolonged cycling due to an irreversible phase transformation from layered to spinel-like structures [7,11–14]. Recently, our group reported that the irreversible phase transformation of Li-rich cathodes could be restrained by stabilization of the Li-rich cathode–electrolyte interface [15]. To resolve these issues, various approaches have been examined, including elemental doping [16–18], coating [19–21], blending materials [22,23], and electrolyte additives [24–30]. Among them, surface modification by electrolyte additives is considered as one of the most efficient because it does not require an additional process compared with other methods. Electrolyte additives tend to readily undergo electrochemical reduction or oxidation at the interface between the electrode and electrolyte to form an artificial protective surface film [10]. Oxidative lithium salt-type additives have recently attracted great attention as promising for the *in situ* formation of a solid electrolyte interphase (SEI) to alleviate the oxidative decomposition of the electrolyte at high voltages. It was reported that the SEI formed by the electrolyte additives magnesium bis(trifluoromethanesulfonyl)imide ($Mg(TFSI)_2$) [31] and lithium bis(fluorosulfonyl)imide (LiFSI) [32] improved the electrochemical performance of high-voltage cathodes. The use of lithium bis(oxalato)borate (LiBOB) [33–36], lithium difluoro(oxalato)borate (LiDFOB) [33–35], lithium tetrafluoro(oxalato)phosphate (LiTFOP) [37–39], and lithium difluoro(bisoxalato)phosphate (LiDFBP) [15] as electrolyte additives also improved the high-temperature cycling performance and rate capabilities of high-voltage cathodes. Additionally, the SEIs formed by the oxidative decomposition of lithium salt-type additives effectively suppressed electrolyte decomposition at high voltages.

In this study, we demonstrate that LiDFOB, used as a salt-type additive, effectively preserves the electrochemical properties of Li-rich cathodes in half-cells and full cells coupled with graphite anodes and allows fast charge transport at the interface between the electrode and electrolyte. In addition, the surface chemistries and morphologies of Li-rich cathodes and graphite anodes cycled in electrolytes with and without 1% LiDFOB were examined by means of *ex situ* X-ray photoelectron spectroscopy (XPS) and scanning electron microscopy (SEM).

2. Experimental

2.1. Preparation of electrolyte and electrode

The $Li_{1.17}Ni_{0.17}Mn_{0.5}Co_{0.17}O_2$, $0.4Li_2MnO_3 \cdot 0.6LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$, cathode material was synthesized according to a procedure reported in the literature [15]. The Li-rich cathode was prepared by spreading a slurry composed of 80 wt% active material ($Li_{1.17}Ni_{0.17}Mn_{0.5}Co_{0.17}O_2$, 10 wt% Super P, and 10 wt% poly(vinylidene fluoride) (PVDF, KF1100, KUREHA Chem. Ind.) dissolved in anhydrous *N*-methyl-2-pyrrolidinone (NMP, 99.5%, Aldrich) on aluminum foil and drying in an oven at 110 °C for 1 h. The dried cathode was pressed with a rolling mill. The thickness, specific capacity, and loading density of the Li-rich cathode were 55 μm , 1.01 mAh cm^{-2} , and 5 mg cm^{-2} , respectively. To remove water adsorbed on the cathodes, they were dried in a vacuum oven at 110 °C for 12 h before cell assembly. The specific capacity of the Li-

rich cathode for a full cell coupled with a graphite anode was 1.442 mAh cm^{-2} . The anode was composed of 96 wt% natural graphite, 1 wt% Super P, and 3 wt% binder (1.5 wt% styrene–butadiene rubber + 1.5 wt% sodium carboxymethyl cellulose). The dried anode was pressed with a rolling mill. The thickness, specific capacity, and loading density of the graphite anode were 71 μm , 1.471 mAh cm^{-2} , and 4.2 mg cm^{-2} , respectively.

A 1.3 M solution of lithium hexafluorophosphate ($LiPF_6$, Soulbrain Co., Ltd.) was prepared in a mixture of ethylene carbonate (EC), ethyl methyl carbonate (EMC), and dimethyl carbonate (DMC) in a 3:4:3 vol ratio. The mixed solvent was treated with calcium hydride (CaH_2) to minimize the water content prior to the addition of the $LiPF_6$ salt to form the baseline electrolyte. The LiDFOB (Soulbrain Co., Ltd.)-containing electrolyte was prepared by adding 1 wt% LiDFOB to the baseline electrolyte. A polyethylene (PE) separator (SK Innovation Co., Ltd.) with a thickness and porosity of 20 μm and 38%, respectively, was used.

2.2. Electrochemical measurements

Electrochemical tests were performed using 2032 coin-type cells assembled in an argon-filled glove box. Half-cells (Li/Li-rich cathode and Li/graphite anode) were galvanostatically precycled at a rate of C/10 between 4.8 V and 2.0, 0.01, and 1.0 V, respectively, using a computer-controlled battery measurement system (Won A Tech WBCS 3000). Thereafter, the cells were cycled at a current density of 120 mA g^{-1} (corresponding to a C/2 rate) over 200 cycles at 25 °C. To demonstrate the impact of LiDFOB on the rate capabilities of Li-rich cathodes and graphite anodes, half-cells were cycled at various discharge C rates: C/2, 1C, 2C, 3C, 4C, 5C, 7C, and 10C.

To examine the effect of LiDFOB on full cell performance, a full cell with a graphite anode and Li-rich cathode was galvanostatically cycled between 4.7 V and 2.0 V in a constant-temperature drying oven (25 °C). Precycling and subsequent cycling were performed at a rate of C/10 and C/2, respectively. After charging to 4.7 V with a constant current, the constant voltage (CV) condition was applied for the precycle; the CV end-condition was determined by a current below C/20.

To investigate the effect of the SEI on the dissolution of transition metal ions from the Li-rich cathodes, fully charged (delithiated) cathodes with baseline- and LiDFOB-derived SEI layers were stored in a polyethylene bottle with the baseline electrolyte at 60 °C for 24 h. The amounts of Mn, Co, and Ni ions in the electrolytes that contacted the fully charged cathodes were measured by means of inductively coupled plasma–mass spectrometry (ICP–MS, ELAN DRC-II).

2.3. Material characterization

For the surface chemistry and morphological analyses of the Li-rich cathodes and graphite anodes, the cells with and without LiDFOB were carefully disassembled in a glove box. Each cathode was rinsed with DMC solvent to remove residual electrolyte and then dried at room temperature. The surface chemistry of the electrodes was examined by *ex situ* X-ray photoelectron spectroscopy (XPS, Scientific K-Alpha system, Thermo Scientific) with Al $K\alpha$ radiation ($h\nu = 1486.6$ eV) under ultrahigh vacuum. XPS spectra were collected using a 0.10 eV step size and 80 eV pass energy. All XPS spectra were energy-calibrated by the hydrocarbon peak at a binding energy of 284.7 eV. The surface morphologies of the electrodes were observed by field-emission scanning electron microscopy (FE-SEM, JSM-6700F, JEOL) in a high-vacuum environment. An energy dispersive spectrometer (EDS) was used to determine the kinds of elements in the region under investigation.

3. Results and discussion

Fig. 1(a) shows the voltage profiles of Li/Li-rich cathode half-cells with and without 1% LiDFOB as a salt-type additive during one precycle at 25 °C. Two voltage plateaus are observed near 4.0 and 4.5 V vs. Li/Li⁺, corresponding to the extraction of Li ions from LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂ and the formation of Li₂O and MnO₂ by the electrochemical activation of Li₂MnO₃, respectively. The cell with the LiDFOB additive displays slightly reduced charge capacity (296.9 mAh g⁻¹) when charged to 4.8 V vs. Li/Li⁺, compared to the baseline electrolyte (302.8 mAh g⁻¹). However, the half-cell with the LiDFOB additive exhibits slightly increased discharge capacity for the Li-rich cathode, indicating the improved initial Coulombic efficiency (ICE) (82.6% → 85.6%). This suggests that the LiDFOB additive inhibits the capacity loss caused by irreversible electrolyte decomposition in the highly oxidized environment at 4.8 V, and thus, leads to a relatively increased discharge capacity (250.2 → 254.4 mAh g⁻¹) despite the slightly reduced charge capacity of the cathode in the presence of LiDFOB. Another noticeable feature is that the voltage drop of the Li-rich cathode with LiDFOB is mitigated during the precycle discharge process. It is thought that the LiDFOB-derived surface film (i.e., the SEI) is less resistive compared with the baseline-electrolyte-derived SEI and thereby allows facile re-lithiation of the Li-rich cathode.

To explore the effect of the LiDFOB-derived SEI on the anodic stability of the electrolyte at high voltages, potentiostatic profiles for Li/Li-rich cathode half-cells with and without 1% LiDFOB at a constant voltage (4.9 V) vs. Li/Li⁺ were obtained, as shown in Fig. 1(b). A comparison of these potentiostatic profiles reveals that

the LiDFOB additive effectively reduces the oxidation currents evolved by electrochemical decomposition of the electrolyte at high voltage. This suggests that LiDFOB contributes to the SEI formation, improving the oxidation stability of the electrolyte. The effect of EC on the anodic stability of electrolytes with and without 1% LiDFOB is given in Fig. S1. In absence of LiDFOB, EC-containing electrolytes led to better anodic stability compared to EC-free electrolytes. Noticeably, the introduction of LiDFOB in EC-containing electrolytes led to excellent anodic stability. From this result, we could confirm

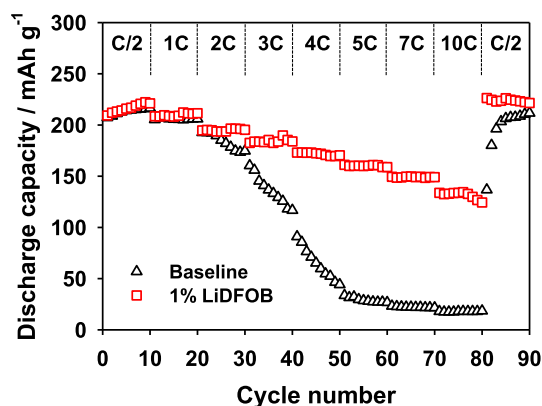


Fig. 2. Rate capability of Li/Li-rich half-cells at various discharge C rates. Charge process was performed at a fixed charge C-rate (C/2).

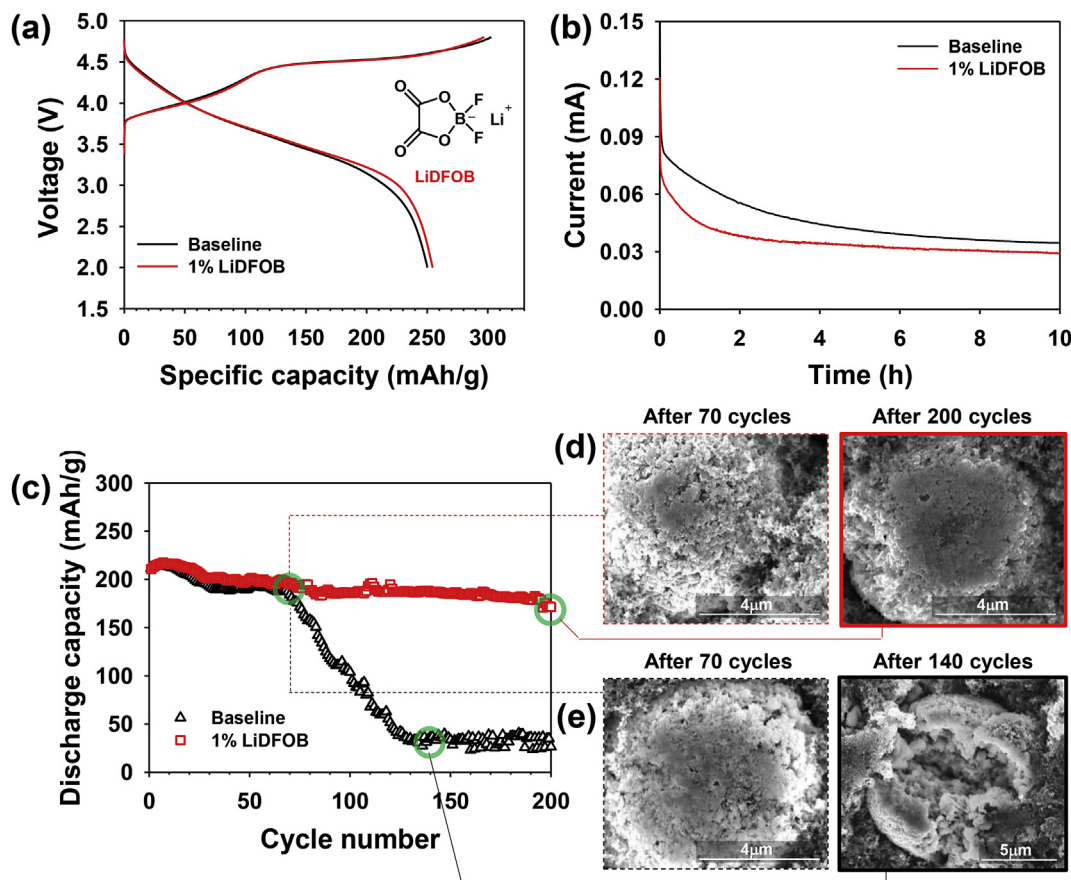


Fig. 1. Electrochemical performance of Li/Li-rich cathode half-cells with baseline and 1% LiDFOB electrolytes at 25 °C: (a) voltage profiles during precycle; (b) potentiostatic profiles at constant voltage of 4.9 V for 10 h after charged to 4.9 V; and (c) discharge capacity retention at a rate of C/2 over 200 cycles. SEM images of Li-rich cathodes with (d) 1% LiDFOB electrolyte at 70th and 200th cycles and (e) baseline electrolyte at 70th and 140th cycles.

that EC-containing electrolytes with 1% LiDFOB builds up robust SEI preventing undesirable electrolyte decomposition in the highly oxidizing environment.

Fig. 1(c) presents the discharge capacities of Li/Li-rich cathode half-cells at 25 °C. The slight increases in the discharge capacities

that appear as cycling begins are mostly due to the activation of the Li_2MnO_3 phase of the Li-rich cathodes [40,41]. The discharge capacity of the Li-rich cathode with the baseline electrolyte shows significant fading after the 70th cycle, and only 12.2% of the initial discharge capacity is delivered after the 200th cycle. Interestingly,

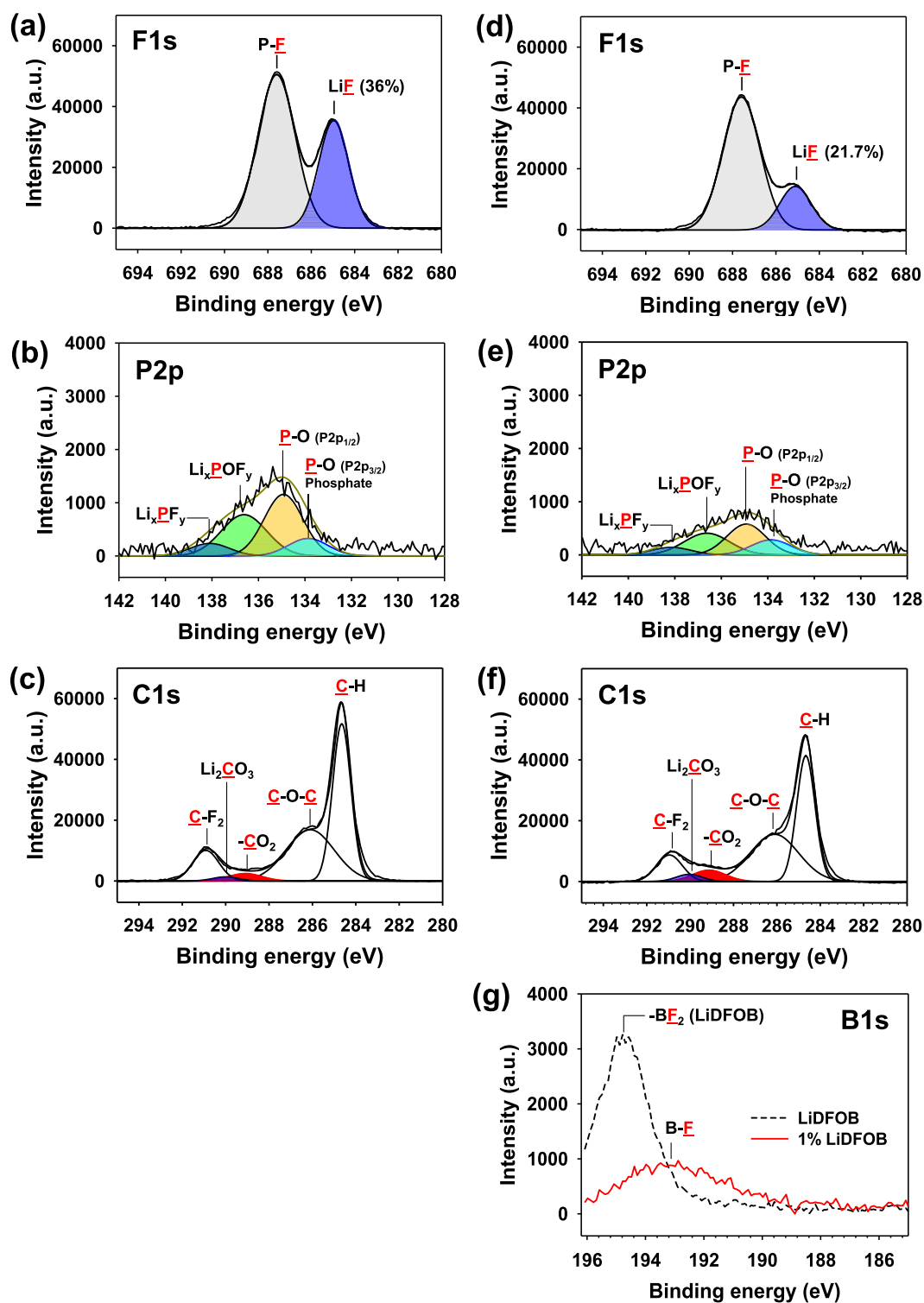


Fig. 3. F 1s, P 2p, and C 1s XPS spectra of Li-rich cathodes after precycle (Baseline: (a), (b), (c); and 1% LiDFOB: (d), (e), (f)). (g) B 1s XPS spectra of Li-rich cathodes precycled in 1% LiDFOB-added electrolyte and LiDFOB.

the introduction of 1% LiDFOB leads to superior cycling stability for the Li-rich cathode, which retains 81.4% of the initial discharge capacity after the 200th cycle.

To determine the origin of the sudden capacity fading of the Li-rich cathode in the baseline electrolyte at the 70th cycle, the morphological changes of the cathodes were investigated (Fig. 1(d) and (e)). Clearly, a striking difference between cathodes cycled in the baseline and 1% LiDFOB-containing electrolytes can be observed. Surprisingly, the Li-rich cathode with 1% LiDFOB maintains its original structure, even after 200 cycles (Fig. 1(d)). However, in the baseline electrolyte, severe microcracking of the Li-rich particles is observed after the 140th cycle, as shown in Fig. 1(e). This finding reveals that the baseline electrolyte is unable to produce a robust SEI layer that can prevent the interfacial degradation of the high-voltage Li-rich cathodes. Consequently, it is thought that the mechanical fracture of Li-rich cathode particles cannot be avoided. Less resistive, uniform SEI layers would be expected to hinder the occurrence of local internal stresses resulting from non-uniform Li extraction and re-lithiation of the cathode materials and prevent the mechanical damage of the cathode. Recently, a degradation mechanism ascribed to non-uniform volume changes of the cathode material during Li extraction and re-lithiation was reported [42].

To elucidate whether the LiDFOB-derived SEI allows fast charge transport at the cathode surface, the rate capability of Li/Li-rich cathode half-cells was examined at high discharge C rates, as shown in Fig. 2. The LiDFOB-derived SEI delivers a high discharge capacity of ca. 131.5 mAh g⁻¹ at a very high current density of 2.4 A g⁻¹ (corresponding to a 10 C rate). In contrast, the Li-rich cathode with the baseline electrolyte undergoes drastic capacity

fading as the C rate increases and a very low discharge capacity of approximately 18 mAh g⁻¹ at the 10 C rate is observed. Thus, the LiDFOB-derived SEI is clearly suitable for facilitating Li-ion transfer in high-voltage Li-rich cathodes.

Fig. 3 shows the surface chemistries of the Li-rich cathodes with and without 1% LiDFOB after the precycle. As clearly shown in the F 1s XPS spectra of Fig. 3(a) and (d), the relative fraction of the LiF peak at 685.0 eV considerably decreases (from 36% to 21.7%) when using 1% LiDFOB. This can be explained by the positive roles of the LiDFOB-derived SEI: 1) it supplies electron-deficient boron atoms that scavenge HF to form resistive LiF [10]; 2) it suppresses LiPF₆ decomposition at the high-voltage Li-rich cathode; and 3) the pathway generating a resistive LiF component is not main mechanism of LiDFOB-derived SEI formation at high voltages. This is in good agreement with the Abraham group's result that reactive oxygen radical intermediate formed by LiDFOB decomposition at high voltages reacts with the EC to produce carbonate-containing polymer species [43]. Since the HF generated by LiPF₆ hydrolysis (LiPF₆ (sol.) + H₂O → POF₃ (sol.) + LiF (s) + 2HF (sol.) and PF₅ (sol.) + H₂O → POF₃ (sol.) + 2HF (sol.)) significantly consumes active Li⁺ ions from the cathode to form resistive LiF as an SEI component, direct contact between HF and the cathode surface should be avoided [10,44,45]. Evidence for the inhibition of LiPF₆ decomposition by the LiDFOB-derived SEI on the cathode is given in the P 2p XPS spectra of Fig. 3(b) and (e). The intensities of the peaks corresponding to Li_xPOF_y, Li_xPF_y, and phosphate (P–O) generated by LiPF₆ decomposition are discernibly reduced for the cathode with 1% LiDFOB. A comparison of the C 1s XPS spectra of the Li-rich cathodes clearly shows that the peak intensity attributed to semi-carbonate-like species (–CO₂–) at 289.1 eV relatively increases

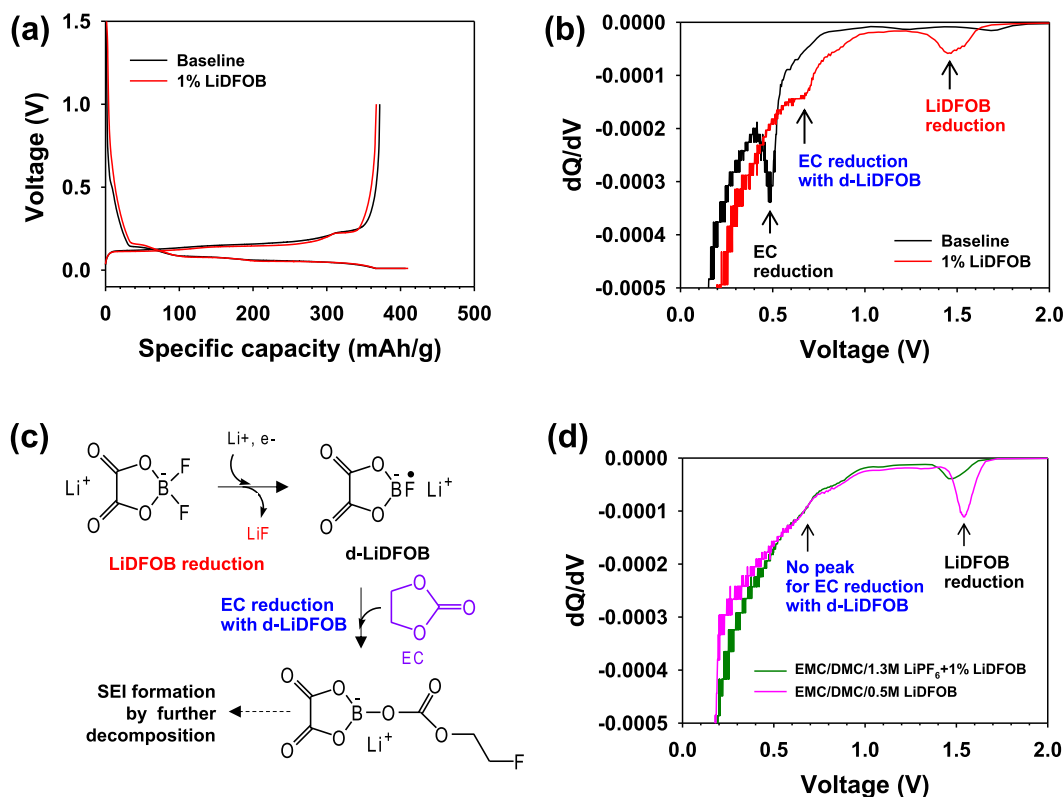


Fig. 4. Electrochemical performance of Li/graphite anode half-cells at 25 °C: (a) voltage profiles during precycle and (b) dQ/dV plots. (c) Possible mechanism for SEI formation by reduction of LiDFOB and EC at the graphite anode. (d) dQ/dV plots of Li/graphite anode half cells with EC-free electrolytes (EMC/DMC/1.3 M LiPF₆ with 1% LiDFOB and EMC/DMC/0.5 M LiDFOB) during first Li intercalation of the precycle.

for the Li-rich cathode with 1% LiDFOB (Fig. 3(c) and (f)). This is probably because decomposition of the oxalate moiety in LiDFOB leads to the formation of SEI components containing $-\text{CO}_2-$ groups on the cathode surface [46]. Note that the peak corresponding to the oxalate group of LiDFOB is located at 289.6 eV. In the B 1s XPS spectra of Fig. 3(g), the Li-rich cathode with 1% LiDFOB exhibits the peak attributed to boron bonded to fluorine (B-F) at 193 eV, which is different with the $-\text{BF}_2$ peak of LiDFOB located at 194.8 eV. This result suggests that compounds with the B-F moiety are produced as SEI components on the cathode.

To confirm the compatibility of the LiDFOB additive with graphite anodes, the electrochemical performance of Li/graphite anode half-cells was examined, as presented in Fig. 4. The graphite anode with 1% LiDFOB displays higher charge (Li intercalation) plateaus and lower discharge plateaus, indicating relatively low polarization compared to that of the baseline electrolyte (Fig. 4(a)). This is because LiDFOB modifies the SEI structure formed by the baseline electrolyte, and thereby, facilitates Li^+ ion transport into/from the graphite. Compared to the baseline electrolyte, which affords a discharge capacity of 371.7 mAh g^{-1} and an ICE of 91.3% during the precycle, the LiDFOB-containing electrolyte delivered a slightly reduced discharge capacity of 367.1 mAh g^{-1} and an ICE of 90.4%. Clearly, the electrochemical reduction of the LiDFOB additive leads to the consumption of Li^+ ions, causing the reversible capacity loss of the graphite during the first Li intercalation.

The dQ/dV plots in Fig. 4(b) provide the reductive decomposition voltages of the baseline and 1% LiDFOB-added electrolytes on the graphite. A relatively high reduction potential near 1.46 V appears for the graphite anode with 1% LiDFOB. It should be noted that the reductive decomposition of LiDFOB takes place prior to the decomposition of EC in the baseline electrolyte [47]. Interestingly, despite the LiDFOB reduction on the graphite, another reduction peak appears near 0.68 V vs. Li/Li^+ . This reduction peak can be explained by a possible reaction between EC and the d-LiDFOB formed by LiDFOB reduction, as depicted in Fig. 4(c). This finding suggests that the reaction between EC and d-LiDFOB can contribute to the increment in peak intensity attributed to semi-carbonate-like species ($-\text{CO}_2-$) in the C 1s XPS spectra of Fig. 3(f). Polar moieties such as $-\text{CO}_2-$ are very effective for ion transport through the SEI. To clarify the possibility of EC reduction by d-LiDFOB at the graphite anode, dQ/dV graphs of graphite anodes precycled with EC-free electrolytes (EMC/DMC/1.3 M LiPF_6 with 1% LiDFOB and EMC/DMC/0.5 M LiDFOB) were obtained, as presented in Fig. 4(d). Notably, the EC-free electrolytes display no reduction peak at $\sim 0.68 \text{ V}$ vs. Li/Li^+ . It is believed that the EC solvent participates in the decomposition reaction pathway of LiDFOB and modifies the surface chemistry of the graphite anode.

The positive impact of the LiDFOB-derived SEI on the cycling performance of the graphite anodes is depicted in Fig. 5. The graphite anode with 1% LiDFOB displays substantially improved cycling stability compared to that with the baseline electrolyte (Fig. 5(a)). The LiDFOB-derived SEI has sufficient ionic permeability at a rate of C/2 as well as enough electrochemical robustness to prevent the irreversible electrolyte decomposition that would cause capacity decay upon repeated charge and discharge processes. Although the use of LiDFOB additive drastically improved cycling performance of the graphite anode with EC-free electrolyte (EMC/DMC/1.3 M LiPF_6), as shown in Fig. 5(b), superior cycling performance of graphite anodes was achieved in EC-containing electrolyte (EC/EMC/DMC/1.3 M LiPF_6) with 1% LiDFOB. It is clear that EC serves as a key component for the formation of LiDFOB-derived SEI to effectively mitigate unwanted electrolyte decomposition at the graphite anode during cycling. The rate capabilities of graphite anodes with and without 1% LiDFOB were evaluated at various discharge rates to understand the impact of the LiDFOB-

derived SEI on the kinetics of charge transfer at the anode-electrolyte interface. The graphite anode with the baseline electrolyte begins to show drastic capacity fading at a mild discharge rate of 1C (Fig. 5(c)). In contrast, no capacity decay appears with the graphite anode with 1% LiDFOB, even at a high discharge rate of 3C. This excellent rate capability of the graphite anode is ascribed to the LiDFOB-derived SEI that may facilitate Li-ion transport at the anode-electrolyte interface.

Fig. 6 shows the F 1s, P 2p, and C 1s XPS spectra obtained from graphite anodes precycled in the baseline and LiDFOB-containing electrolytes. As seen in the F 1s XPS spectra in Figs. 3(a),(d) and 6(a),(d), the peak intensity assigned to LiF at 685.0 eV is considerably increased for the graphite anode with 1% LiDFOB, unlike the Li-rich cathodes with 1% LiDFOB, which show a reduced LiF fraction. This confirms that the LiDFOB additive leads to different surface chemistries (affording a LiF-less SEI for the Li-rich cathode and a LiF-rich SEI for the graphite anode). The highly improved rate capability of the graphite anode obtained with the LiDFOB additive

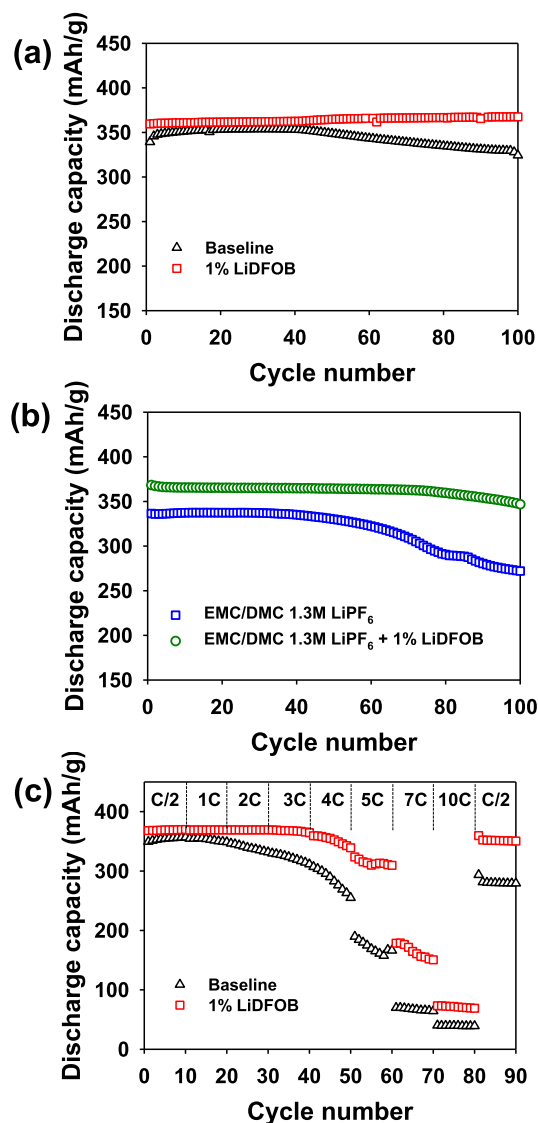


Fig. 5. Discharge capacity retention at a rate of C/2 over 100 cycles: (a) EC-containing electrolytes with and without 1% LiDFOB and (b) EC-free electrolytes with and without 1% LiDFOB. (c) Rate capability at various discharge C rates. Charge was performed at a fixed charge C-rate (C/2).

can be explained by the reduced relative amounts of meta-stable products (Li_xPF_y , Li_xPOF_y , P-O) formed by LiPF_6 decomposition and the appearance of polar moieties ($-\text{CO}_2-$) in the SEI created in the presence of LiDFOB (Fig. 6(b)–(f)) [48]. The reduced fraction of LiPF_6 decomposition products and increased proportion of LiF in

the SEI on the graphite surface indicate that the reductive decomposition of LiPF_6 leads to the formation of LiF . It is thought that LiF can suppress the formation of the resistive SEI by undesirable decomposition of the electrolyte unlike the meta-stable products (Li_xPF_y , Li_xPOF_y , P-O) generated by the LiPF_6 decomposition, and

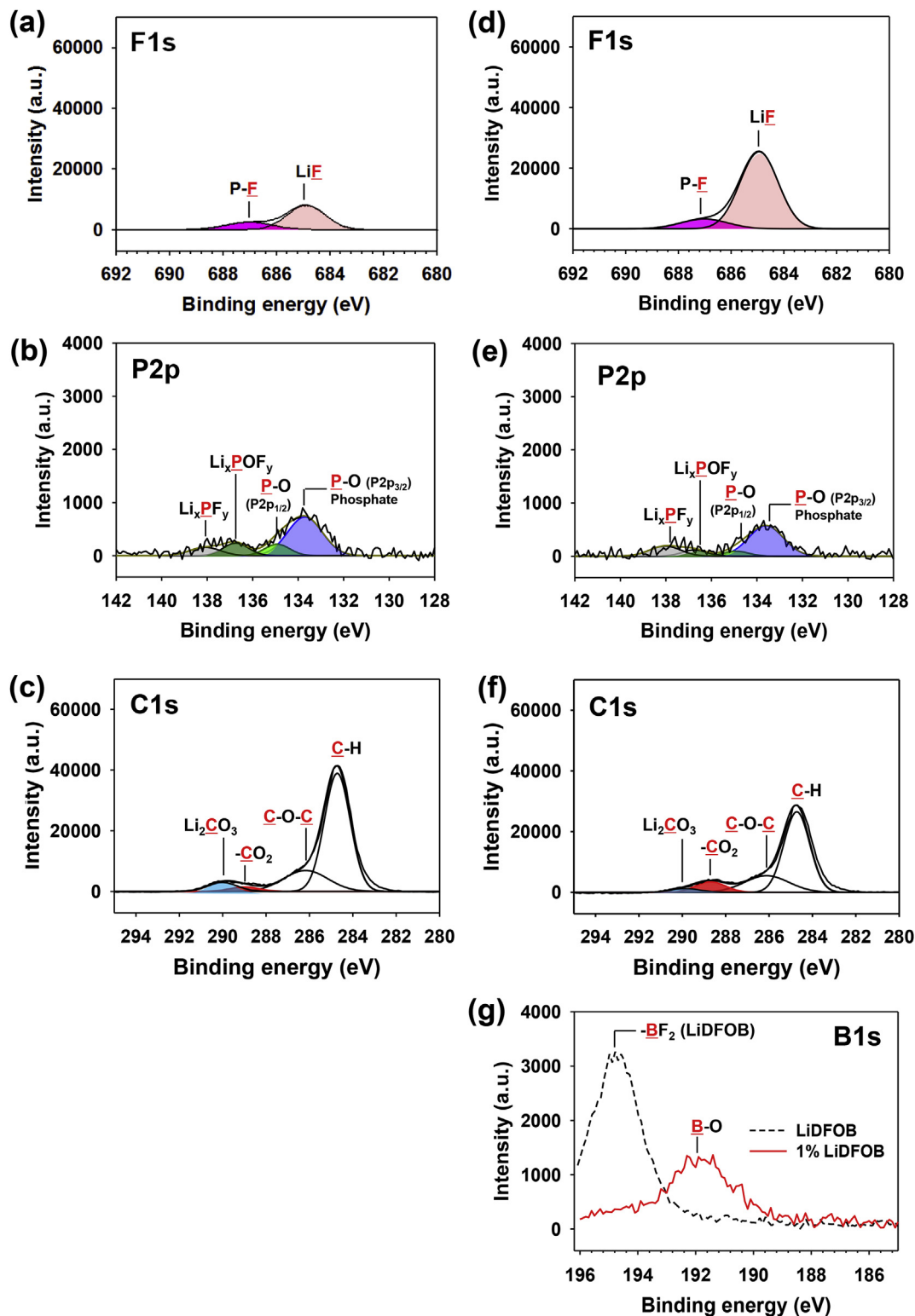


Fig. 6. F 1s, P 2p, and C 1s XPS spectra of graphite anodes after precycle (Baseline: (a), (b), (c) and 1% LiDFOB : (d), (e), (f)). (g) B 1s XPS spectra of graphite anodes precycled in 1% LiDFOB -added electrolyte and LiDFOB .

thereby, the graphite anodes with 1% LiDFOB deliver high discharge capacities at high discharge C rates. In addition, polar groups, including $-\text{CO}_2-$, can interact with Li ions via ion-dipole interactions and facilitate fast ion transport through the SEI layer.

A noticeable feature for the graphite anode with 1% LiDFOB is the appearance of boron bonded to oxygen (B-O) at 191.8 eV that are not observed in the SEI on the Li-rich cathode. It can be thought that the disappearance of the peak corresponding to B-F is mostly due to the LiF formation by the reaction between Li and BF_2 group of LiDFOB.

The cycling performance of full cells constructed with graphite anodes, Li-rich cathodes, and the electrolytes with or without 1% LiDFOB is shown in Fig. 7. Slightly increased initial Coulombic efficiencies and discharge capacities are achieved for the full cells with the 1% LiDFOB electrolyte. It is likely that the salt-type LiDFOB additive forms desirable SEIs on the graphite anode and Li-rich cathode in the full cell. Further evidence for the beneficial functions of LiDFOB, which can impart interfacial stabilization for both electrodes, is given in Fig. 7(b). The discharge capacity retention at the 100th cycle is substantially improved for the cell with 1% LiDFOB (82.7%) compared to that with the baseline electrolyte (45.8%). After 200 cycles, a discharge capacity retention of only 11% is obtained for the full cell with the baseline electrolyte, whereas the introduction of 1% LiDFOB avoids severe capacity fading of the full

cell. Moreover, the Li-rich/graphite full cell cycled in the electrolyte with 1% LiDFOB exhibits a discernable Coulombic efficiency improvement of greater than 99.5% over 100 cycles at 25 °C (Fig. 7(b)).

To understand the significant capacity fading of the Li-rich/graphite full cell with the baseline electrolyte after 70 cycles, graphite anodes retrieved from these cells after 200 cycles were analyzed by SEM and EDS. Unlike pristine graphite (Fig. 7(c) and (f)), which exhibits a very clean surface, the graphite anode from the full cell cycled 200 times in the baseline electrolyte has a granular surface (Fig. 7(d) and (g)). Peaks assigned to Co, Mn, and Ni, which are not observed for the pristine graphite, appear. It is thus believed that the baseline electrolyte does not form a robust SEI that can suppress metal-ion dissolution from the Li-rich cathode or metal deposition on the graphite anode. Since the deposition of these metals (Co, Mn, and Ni) causes the consumption of electrons necessary for the intercalation of Li^+ ions into the graphite, considerable capacity fading of the full cells with their limited Li^+ sources cannot be avoided.

Furthermore, the uneven plating of the metallic Mn and Ni on the graphite anode leads to further electrolyte decomposition, and a non-uniform SEI showing a granular surface morphology is produced. In contrast, with the LiDFOB-modified surface chemistries of both electrodes in the full cell, Co, Mn, and Ni signals in the EDS

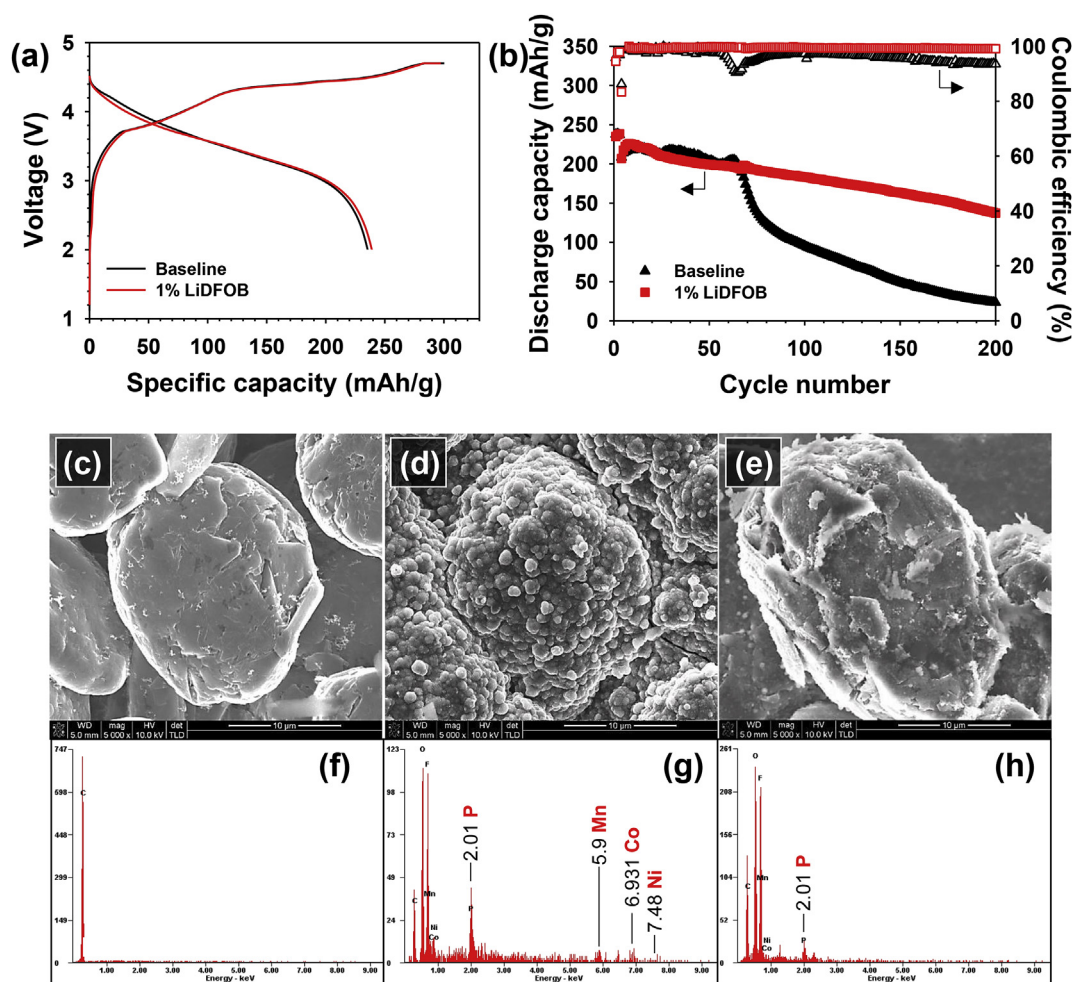


Fig. 7. Electrochemical performance of Li-rich/graphite full cells at 25 °C: (a) voltage profiles during precycle; and (b) discharge capacity retention and Coulombic efficiency at a rate of C/2 over 200 cycles. SEM images and EDS patterns of (c,f) pristine (non-cycled) graphite anode and graphite anodes retrieved from full cells with (d,g) baseline and (e,h) 1% LiDFOB electrolytes after 200 cycles.

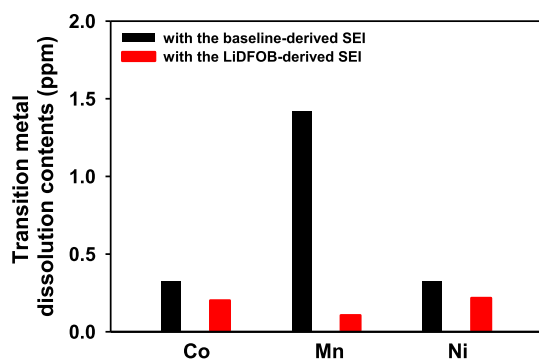


Fig. 8. A comparison of transition metal dissolution (Co, Mn, Ni) from fully delithiated Li-rich cathodes with the SEI layer derived from decomposition of the baseline electrolyte or 1% LiDFOB-added electrolyte at 60 °C for 24 h.

pattern of the graphite anode are not observed, and a relatively smooth anode surface is obtained after cycling (Fig. 7(e) and (h)). These results clearly show that the LiDFOB-derived SEI on the electrodes is sufficiently robust to mitigate metal-ion dissolution from the Li-rich cathode and avoid metal deposition on the graphite anode.

To verify the effect of the LiDFOB-derived SEI on metal-ion dissolution from Li-rich cathodes, ICP-MS analysis was performed for fully delithiated cathodes with baseline- or LiDFOB-derived SEIs. As seen in Fig. 8, the extent of metal (Co, Mn, and Ni) ion dissolution from a fully delithiated cathode after 24 h at 60 °C is drastically reduced by the presence of the LiDFOB-derived SEI on the cathode surface. It is obvious that the LiDFOB-derived SEI on the cathode acts as a more resistive layer toward metal-ion dissolution compared to the baseline-derived SEI. Small amounts of metal ions in the electrolyte are detrimental for battery performance because metal deposits are formed by consuming the electrons for the intercalation of Li^+ ions into the graphite, and metallic elements (Co, Mn, and Ni) can lead to the non-uniform growth of unstable SEI layers that hamper charge transfer reactions at the graphite anode. Therefore, it can be thought that the LiDFOB-derived SEI is thermally robust and preserves the electrochemical performance of both electrodes in a full cell.

4. Conclusions

We have demonstrated that LiDFOB as a salt-type additive improves the oxidation stability of the electrolyte under high voltage conditions and alleviates metal-ion dissolution from Li-rich cathodes at high temperature. The XPS spectral study confirmed that the use of LiDFOB additive beneficially modifies the surface chemistries of both the graphite anode and Li-rich cathode, and revealed a reductive decomposition mechanism for the EC component of the electrolyte that is involved in the formation of the LiDFOB-derived SEI. Furthermore, the LiDFOB-derived SEI provides efficient electronic and ionic transport pathways at the electrode-electrolyte interface and realizes a substantial improvement in the cycling performance of a high-voltage Li-rich cathode coupled with a graphite anode.

Acknowledgements

This work was supported by the IT R&D program of the Ministry of Trade, Industry and Energy/Korea Evaluation Institute of Industrial Technology (MOTIE/KEIT) (Development of Li-rich Cathode and Carbon-free Anode Materials for High Capacity/High Rate Lithium Secondary Batteries, 10046306). This research was partly

supported by a National Research Foundation of Korea Grant funded by the Korean Government (MEST) (NRF-2013-C1AAA001-0030538).

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2017.04.094>.

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